Neel Gali

Software Engineer

neelgali7@gmail.com | +1(475)-291-1455 | Austin, TX

Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Epik Solutions | AI-Powered Physical Security Platform

- Developed an **AI-powered platform** for **video security** using **TensorFlow** and **Keras**, enabling real-time anomaly detection and reducing false positives by 30% across 500+ cameras.
- Implemented a **binary classifier** for **object detection** in camera feeds using **C++** and **OpenCV**, achieving 95% accuracy in differentiating states for access control.
- Engineered scalable services for **unusual object addition/removal** detection utilizing **Python** and **AWS Lambda**, processing 10,000 events/second with 100ms latency.
- Designed and optimized **deep learning networks** for **people frequencies** analysis, improving crowd density estimates by 25% for critical safety monitoring systems.
- Architected robust **data structures** for multi-sensor data fusion, supporting real-time processing across **video security** and **air quality sensors**, ensuring data integrity.
- Led **agile team** software development for critical **access control** features, delivering new functionalities 15% faster through continuous integration practices and rigorous testing.
- Developed **modular C++ code** for embedded systems controlling **alarms** and **intercoms**, enhancing device reliability by 18% and reducing firmware update times.

Software Engineer *Jul 2022 – Jun 2025*

Samay Systems LLC | Enterprise AI & Data Solutions

- Developed **Python** services for **object counting** in security footage, integrating **OpenCV** and **AWS S3** to process 100GB daily, reducing manual effort by 40%.
- Contributed to **deep learning networks** for anomaly detection using **TensorFlow** on **GCP AI Platform**, decreasing false positives in sensor data by 22%.
- Implemented **data structures** for efficient storage and retrieval of time-series sensor data, optimizing query performance by 35% using **PostgreSQL** on **AWS RDS**.
- Designed and built **REST APIs** using **Python Flask** for secure data ingestion, supporting 5000+ daily requests from **access control** devices with 99.9% uptime.
- Automated infrastructure deployment for ML model training environments using **Docker** and **Kubernetes** on **GCP**, cutting setup time from hours to minutes.
- Utilized **traditional computer vision algorithms** for image pre-processing and feature extraction, improving subsequent **neural net frameworks** performance by 15%.
- Collaborated in an **agile team** environment to develop and deploy a predictive maintenance system, reducing equipment downtime by 10% through proactive alerts.
- Developed real-time monitoring dashboards using **Grafana** and **Prometheus** for **AI-powered platform** health, improving incident response time by 20% across 20+ services.

Software Engineer *Jul 2020 – Aug 2022*

Zemoso Technologies | Cloud-Native Application Development

- Developed backend services using **Java Spring Boot** and **PostgreSQL** on **AWS EC2**, handling 1M daily transactions for a B2B SaaS platform.
- Implemented robust **REST APIs** for client-facing applications, ensuring data security and integrity, which processed 500,000 requests per day efficiently for users.
- Optimized database queries and **data structures** for improved application performance, reducing report generation time by 40% for financial analytics.
- Deployed microservices into **AWS ECS** using **Docker** containers, achieving 99.9% uptime and scaling efficiently to support increased user load.
- Collaborated with an **agile team** to develop new features for an enterprise resource planning system, increasing user engagement by 15% through rapid iterations.
- Contributed to design and implementation of system **architecture** for a new module, reducing development time by 20% using established design patterns.

Skills

Languages: Python, C++, Java
Frameworks & Tools: TensorFlow, Keras, OpenCV, REST APIs, Flask, Docker, Kubernetes, Spring Boot, Grafana, Prometheus, OpenAI API
Cloud & Infra: AWS, GCP, AWS Lambda, AWS S3, AWS RDS, AWS EC2, AWS ECS, PostgreSQL
Concepts: Deep Learning, Computer Vision, Data Structures, System Architecture, Agile, Microservices

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India